

Analysis and Design of a Low-Voltage Low-Power Double-Tail Comparator

A Project Report Submitted

in Partial Fulfilment of the Requirements

for the Degree of

MASTER OF TECHNOLOGY

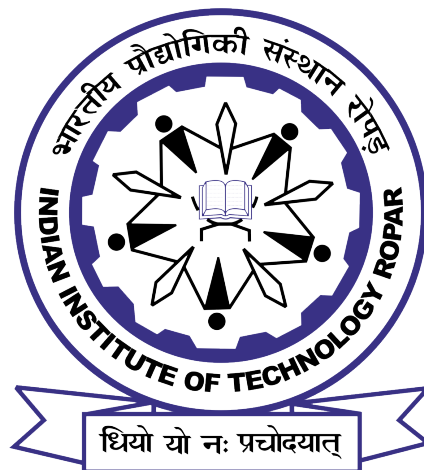
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April, 2026

Declaration

In this regard, I certify that the following project report “**Analysis Design of a Low Voltage Low Power Double Tail Comparator**” is a product of my hard work undertaken under the supervision of Prof. Devarshi Mrinal Das. None of the facts stated in the report have been borrowed from other sources; they are true and authentic. The report is an original document and does not include any form of plagiarism.

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Acknowledgement

I would like to express my sincere gratitude to my guide, Prof. Devarshi Mrinal Das, for his support during this study. I would also like to thank the Department of Electrical Engineering, IIT Ropar, for providing all the required facilities. Finally, I am thankful to my family members and friends for their encouragement.

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Certificate

This is to certify that the project report titled “**Analysis and Design of a Low-Voltage Low-Power Double-Tail Comparator**” submitted by **Prashant Kumar Gautam (2025EEM1004)** is an original piece of research carried out under my supervision. As far as I am aware, no part of the research has been submitted to any other institution in support of a qualification elsewhere.

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Abstract

Fast, low-power and small size comparators have always been needed for modern ADC circuits. This paper will explore the behavior of dynamic comparators with special emphasis on their delay and power dissipation. Both conventional and double tail comparators will be considered to identify how they work and how they fail. From this analysis, an enhanced double tail configuration will be suggested to enhance the performance at low supply voltages. A circuit is designed and simulated on Cadence Virtuoso based on 0.18 micron CMOS process. The simulation shows better speed and reduced power consumption compared to existing designs.

Index Terms— Dynamic comparator, double-tail comparator, low-power design, high-speed ADC.

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Chapter 1

Introduction

1.1 Overview of ADCs

The Analog-to-Digital Converter (ADC) forms the basis of electronics today, providing conversion from a continuous analog signal into a digital form for further processing. These converters abound in modern-day electronic circuits including communication circuits, instrumentation devices, and even systems used for data acquisition.

Performance parameters of an ADC typically include speed of operation, resolution capability, power consumption, and linearity. With advancement in complementary metal oxide semiconductor (CMOS) technology leading to reduced transistor sizes, the demand for low power/high speed ADCs is ever increasing. [2].

1.2 Role of Comparator in ADC

At the core of most ADC designs are comparators, particularly those that are used in high-speed ADC types such as flash and successive approximation. The main function of this comparator is to compare the sampled signal to the reference value and produce a digital output based on which of the two is higher.

The performance of this comparator component directly influences the efficiency of the ADC as a whole. Important parameters, including the decision time, input referred offset, and power consumption, will ultimately determine the quality of the signal conversion process. Therefore, designing an efficient comparator will play a key role in developing ADCs with good performance [3].

1.3 Need for Low-Power High-Speed Comparators

With continuous advancement in CMOS technology, reduction in supply voltages is continuously being done in order to reduce power consumption. However, when supply voltage is decreased, designing a fast comparator becomes difficult due to certain reasons.

The reduction in voltage results in the reduction of transistor transconductance and headroom, and thus, reduction in performance. Thus, for achieving both high speed and reduced power consumption, intelligent tradeoffs need to be made while designing.

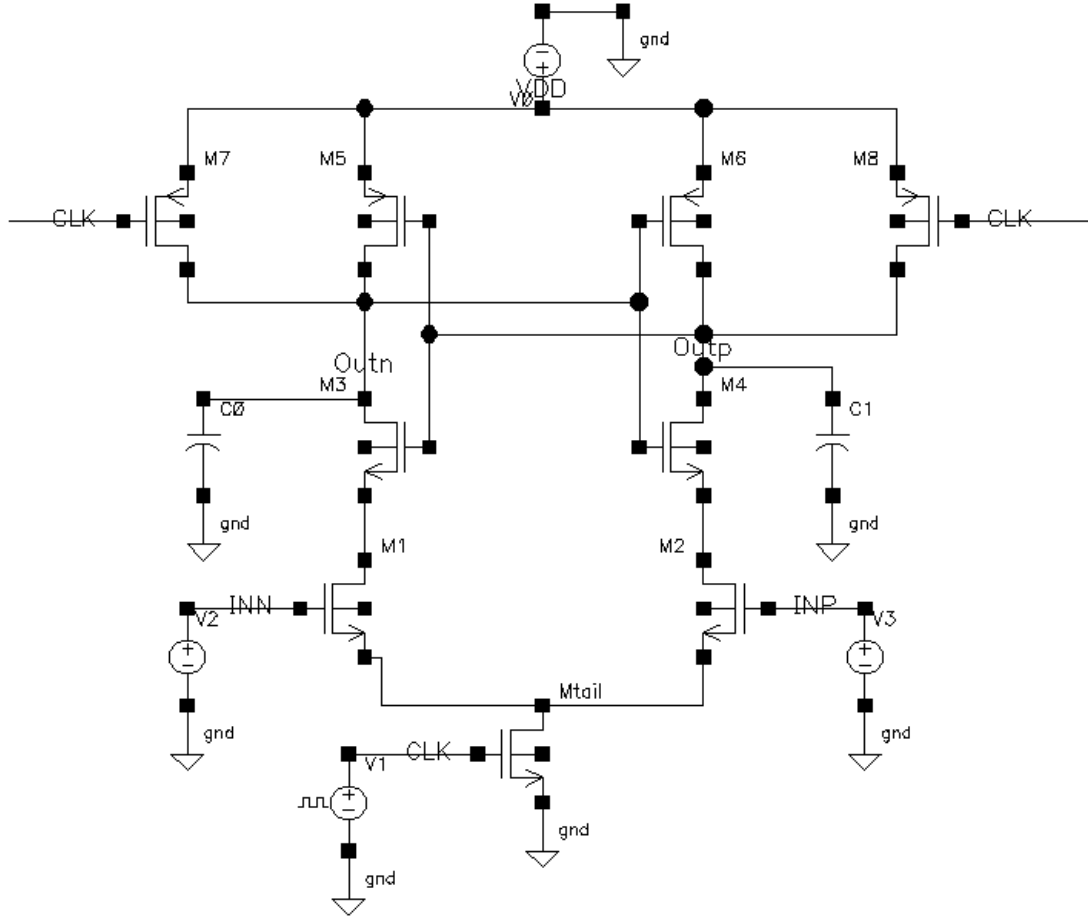


Figure 1.1: Schematic diagram of a conventional dynamic comparator

Dynamic regenerative comparator is one such design that allows high-speed operation along with zero static power consumption.

1.4 Problem Statement

A standard dynamic comparator has high speed and low power dissipation in a static state. However, it cannot operate satisfactorily at lower supply voltages as the transistors are stacked further, and headroom gets reduced, thus resulting in sensing delay.

The double-tail dynamic comparator is an attempt to solve this problem by maintaining isolation between the input stage and the latch circuit, which makes it suitable for low-voltage operation. Nevertheless, a new delay occurs due to extended sensing time for smaller input voltages and increased parasitic capacitive load.

Hence, a need for better comparator circuits that can be operated under low-voltage conditions is obvious.

1.5 Objective of the Work

This project focuses on designing a low voltage, low power double tail dynamic comparator for high-speed analog-to-digital converters.

We analyze the conventional comparator architectures and double tail comparator topologies to identify their limitations. Based on our analysis, we proceed to design an enhanced comparator circuit that would be faster and consume less power.

Our novel comparator is designed and tested using Cadence Virtuoso tool with 0.18um CMOS technology. Various key parameters such as propagation delay, power dissipation, and speed of operation are evaluated through simulations.

Chapter 2

Literature Review

2.1 Introduction

In ADC systems, comparators are extremely important components, particularly if you want to achieve high speeds of operation. At such speeds of operation, these comparators need to make an instant decision, which means that delay times, power consumption, and offset voltages become highly significant [2, 3].

There are many types of comparators available but one type of particular interest is clocked regenerative comparator, which operates at very high speeds without dissipating any significant amount of power. Numerous research efforts have been directed toward developing fast and low-power comparators [4, 5].

2.2 Clocked Regenerative Comparators

Clocked regenerative comparators use feedback to amplify small differences into digital signals. Such comparators usually operate in two phases; these include reset phase and evaluation phase [2].

In the reset phase, the comparator voltages are adjusted to a known value. In the evaluation phase, the comparator amplifies its signal to give definite digital output [3, 4].

2.2.1 Conventional Dynamic Comparator

The conventional dynamic comparator is widely used due to its simple structure and zero static power consumption. During the reset phase, output nodes are precharged to the supply voltage, while in the evaluation phase they discharge depending on the input voltage difference [2].

The delay of the comparator consists of two components : discharge delay (t_0) and latch delay (t_{latch}).

Discharge Delay :

$$t_0 = \frac{C_L \cdot V_{thp}}{I/2} \approx \frac{2C_L V_{thp}}{I_{tail}}$$

Latch Delay :

$$t_{latch} = \frac{C_L}{g_{m,eff}} \ln \left(\frac{V_{DD}}{2V_0} \right)$$

Total Delay :

$$t_{delay} = t_0 + t_{latch}$$

From the delay equations, it is clear that the total delay depends on load capacitance, tail current, effective transconductance, and the initial voltage difference. Increasing the tail current or transconductance reduces delay, while larger capacitance increases delay [6].

Moreover, increasing the initial voltage difference (ΔV_0) significantly reduces the latch delay due to the logarithmic relationship.

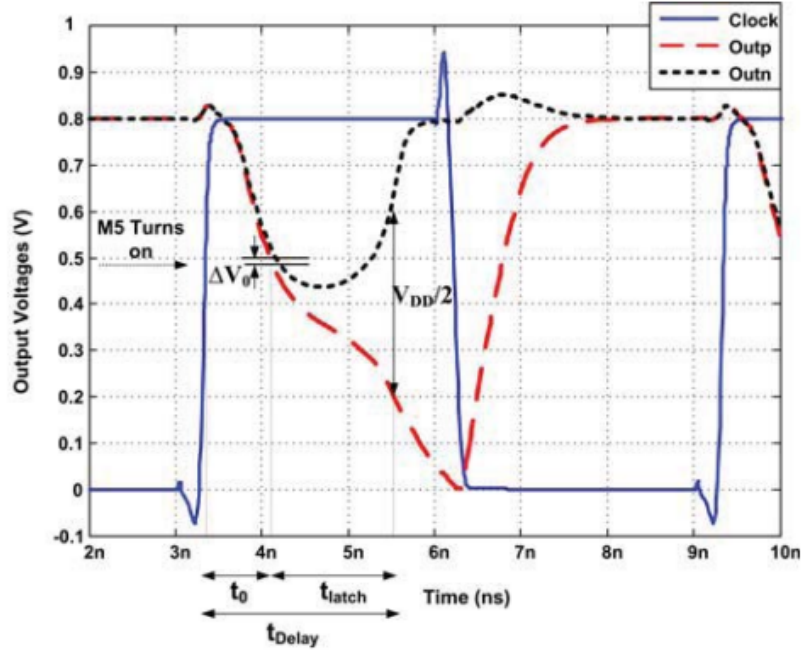


Figure 2.1: Transient response of conventional dynamic comparator as reported in [1]

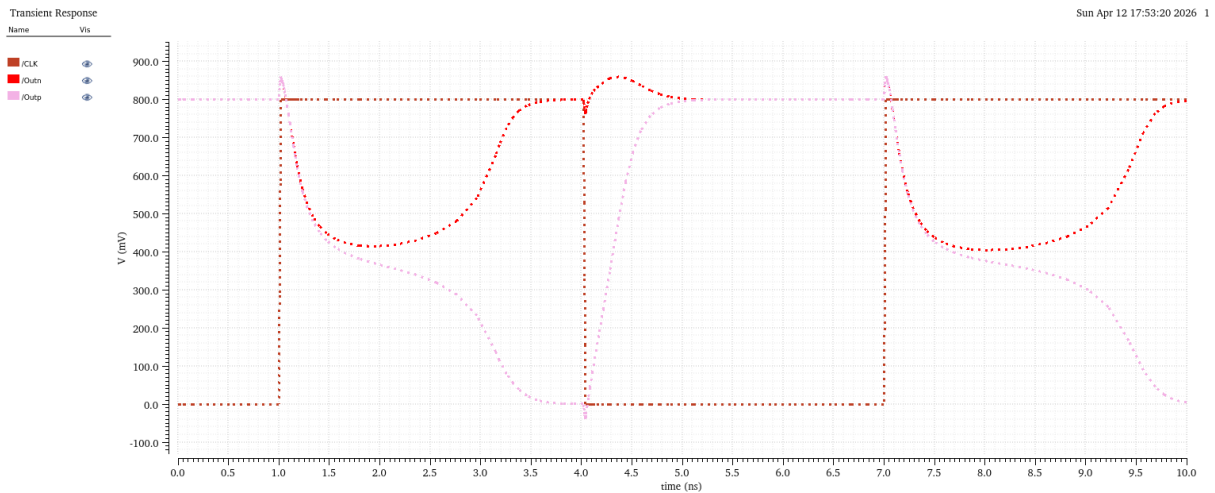


Figure 2.2: Transient response of conventional dynamic comparator obtained using Cadence

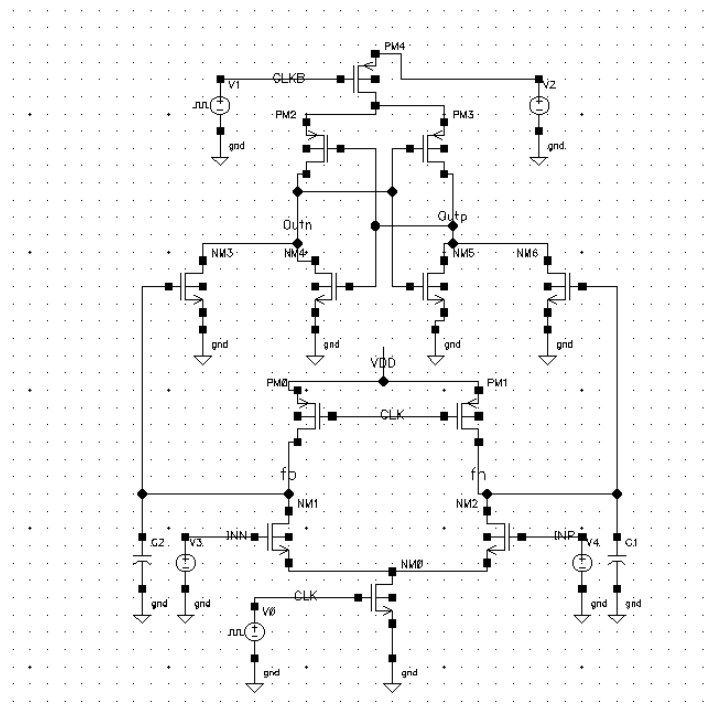
Figure 2.2 shows the transient response of the conventional dynamic comparator obtained through Cadence Virtuoso [7]. The clock signal controls the operation of the comparator,

During the reset phase, the output nodes are precharged to the supply voltage level. When the clock signal switches to the evaluation phase, the comparator begins to respond to the input differential signal, causing the output nodes to discharge at different rates [2, 4].

The total propagation delay of the comparator can be divided into two categories: the first one is the discharge delay, while the second one is the regeneration delay. If the output signal exceeds the halfway point of the supply voltage, positive feedback helps to switch faster

2.2.2 Conventional Double-Tail Dynamic Comparator

Having two tail transistors makes sure that both input and latch currents are independently controllable, hence improving speed and minimizing offsets.



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Discharge Delay :

$$t_0 = \frac{V_{thn} C_{Lout}}{I_{B1}} \approx \frac{2 V_{thn} C_{Lout}}{I_{tail2}}$$

Initial Voltage Difference :

$$V_0 = \frac{2 V_{thn} g_{mR1,2}}{g_{m1,2}} \cdot \frac{V_{in}}{I_{tail2}/2}$$

Total Delay :

$$t_{delay} = t_0 + \frac{C_{Lout}}{g_{m,eff}} \ln \left(\frac{V_{DD}}{V_0} \right)$$

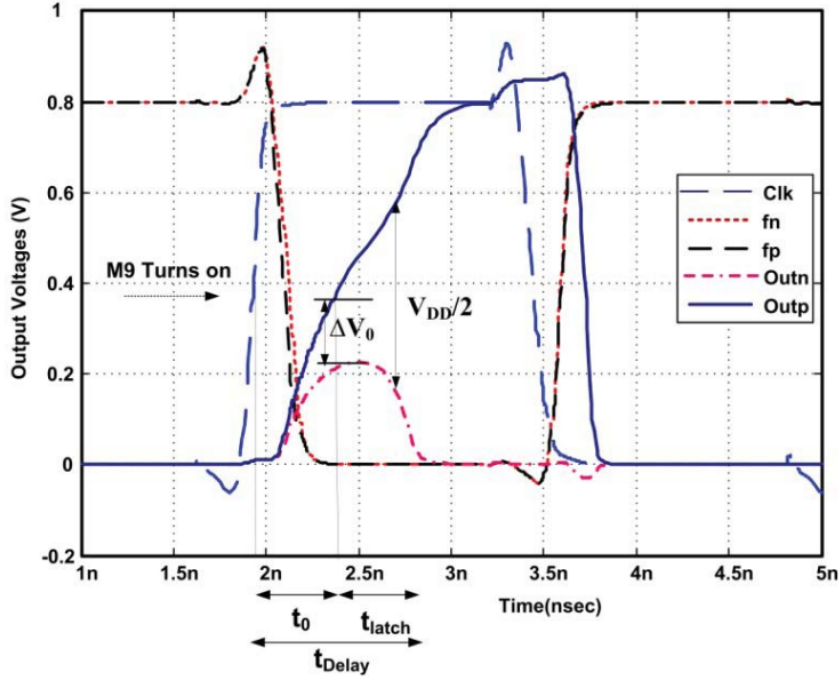


Figure 2.4: Transient response of double-tail dynamic comparator as reported in [1]

Figure 2.4 shows the transient response of the conventional double-tail dynamic comparator. During the reset phase, internal nodes are precharged to the supply voltage, while in the evaluation phase they begin to discharge based on the input difference.

A small initial voltage difference (ΔV_0) is developed between the internal nodes, which is amplified by the latch through positive feedback. The total delay consists of discharge delay (t_0) and latch delay (t_{latch}).

Once the output crosses ($V_{DD}/2$), strong regeneration occurs, leading to a rapid transition to full logic levels.

Figure 2.5 shows the transient response of the double-tail comparator obtained using Cadence Virtuoso simulation. The clock signal controls the switching between reset and evaluation phases.

During the evaluation phase, the intermediate nodes (f_n) and (f_p) begin to discharge based on the input differential voltage. A small initial voltage difference (ΔV_0) is generated between these

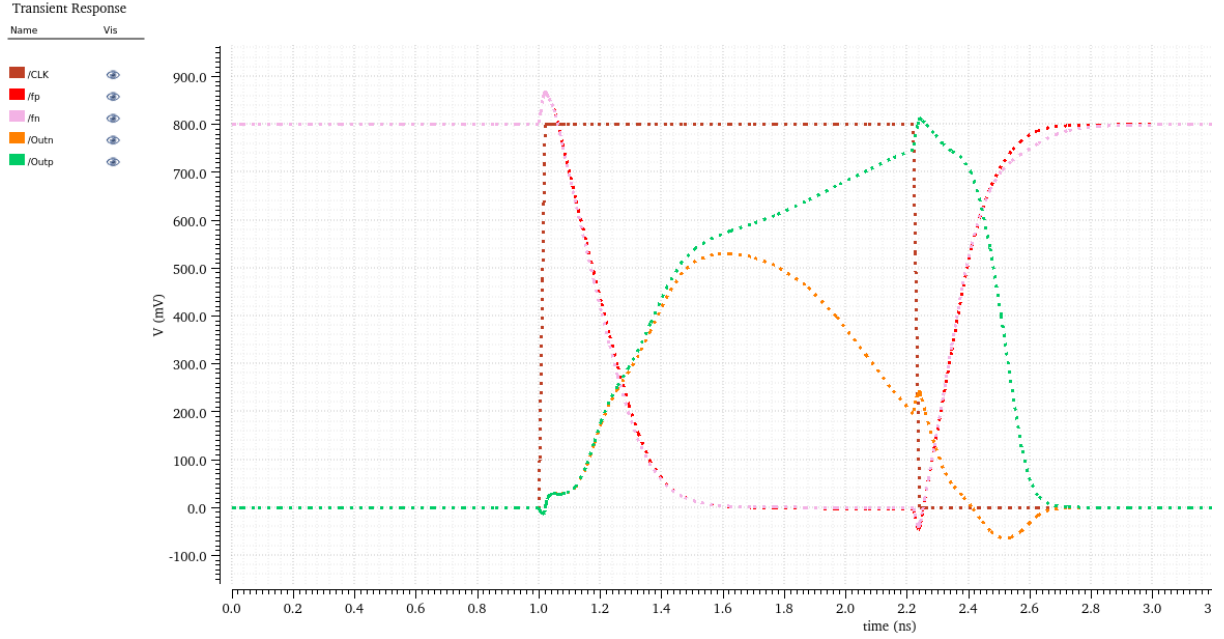


Figure 2.5: Transient response of double-tail dynamic comparator obtained using Cadence

nodes.

This voltage difference is transferred to the output nodes, where the cross-coupled latch amplifies it through positive feedback. The outputs rapidly transition to logic levels once the voltage crosses $(V_{DD}/2)$, indicating successful comparison.

The double-tailed comparator outperforms the traditional dynamic comparator at low voltages. This is possible since this comparator reduces stacking of transistors and steers the current in a more efficient manner.

It can be seen from Fig. 2.4 and Fig. 2.5 that the waveforms of the transients are similar, and hence the operation of the comparator is as expected. The waveform represents the discharge stage, an increasing initial difference of the voltages, and the rapid regeneration process.

At low supply voltages, however, the double-tailed comparator performs better due to reduced stacking and better steering of the currents.

2.3 Limitations of Existing Comparators and Need for Proposed Design

2.3.1 Limitations of Conventional Dynamic Comparator:

The traditional dynamic comparator suffers inefficiencies with decreasing supply voltages due to transistor stacking reducing headroom, leading to increased delays and reduced sensitivity [2].

Likewise, the double-tailed comparator performs better at low-voltage conditions but still has its drawbacks: it consumes more power and regeneration is slower since the initial voltage

differences are relatively small [4, 5].

2.3.2 Limitations of Double-Tail Dynamic Comparator:

In spite of all these benefits, the double-tailed comparator falls short due to certain limitations.

Firstly, due to the intermediate nodes that the comparator contains, there arise additional parasitic capacitances, thus leading to an increased power consumption because of charging/discharging operations.

Secondly, the comparator suffers from lack of speed, since the operation time depends on the creation of the differential voltage at the internal nodes. At the beginning of the evaluation phase, the difference (ΔV_0) is very small, which slows down the regeneration process.

Thus, although the new structure has advantages over the conventional one, its speed and power performance have not been maximized. [1].

2.3.3 Need for Proposed Comparator:

For better performance, a better comparator structure is required.

The recommended structure employs double tail comparator which focuses on the regeneration stage. It amplifies the positive feedback process, thus improving the effective transconductance as well as the voltage difference (ΔV_0).

This implies that quicker decision making and delay reductions are achieved at minimal power consumption and without increasing the complexity of the circuit.

Overall, it can be seen that the new comparator offers an optimal compromise between speed, power, and low-voltage requirements [1].

Chapter 3

Methodology

3.1 Introduction

This section will cover the design and simulation of the low voltage, low power double tail comparator. The conventional comparator suffers from many drawbacks where it is challenging to obtain high speed at the expense of power consumption. To overcome such problems, changes were introduced to the double tail circuit to eliminate such limitations.

The complete circuit was fabricated and tested via Cadence Virtuoso using the 0.18um CMOS technology. This paper will cover all the operations of the circuit and its design steps.

3.2 Proposed Comparator Architecture

The topology of the comparator remains in the conventional two tail form, though extra transistors are added as part of the control function [1].

In essence, what is maintained is the traditional two tail form, but supplemented with extra transistors that will help increase the initial voltage differential between the internal nodes.

This can be achieved by incorporating cross-coupling control transistors, which will cause a quick increase in the voltage difference between the internal nodes. In addition, switching transistors are utilized to block extraneous paths for the current [1].

3.3 Circuit Diagram of the Proposed Comparator

Figure 3.1 illustrates a schematic diagram of the designed comparator using Cadence Virtuoso software. The components used include the input differential pair, the latch stage, the control transistors, and the switching transistors.

The additional control transistors are vital since they help to increase the voltage across the internal nodes, thus improving regeneration time and reducing delays.

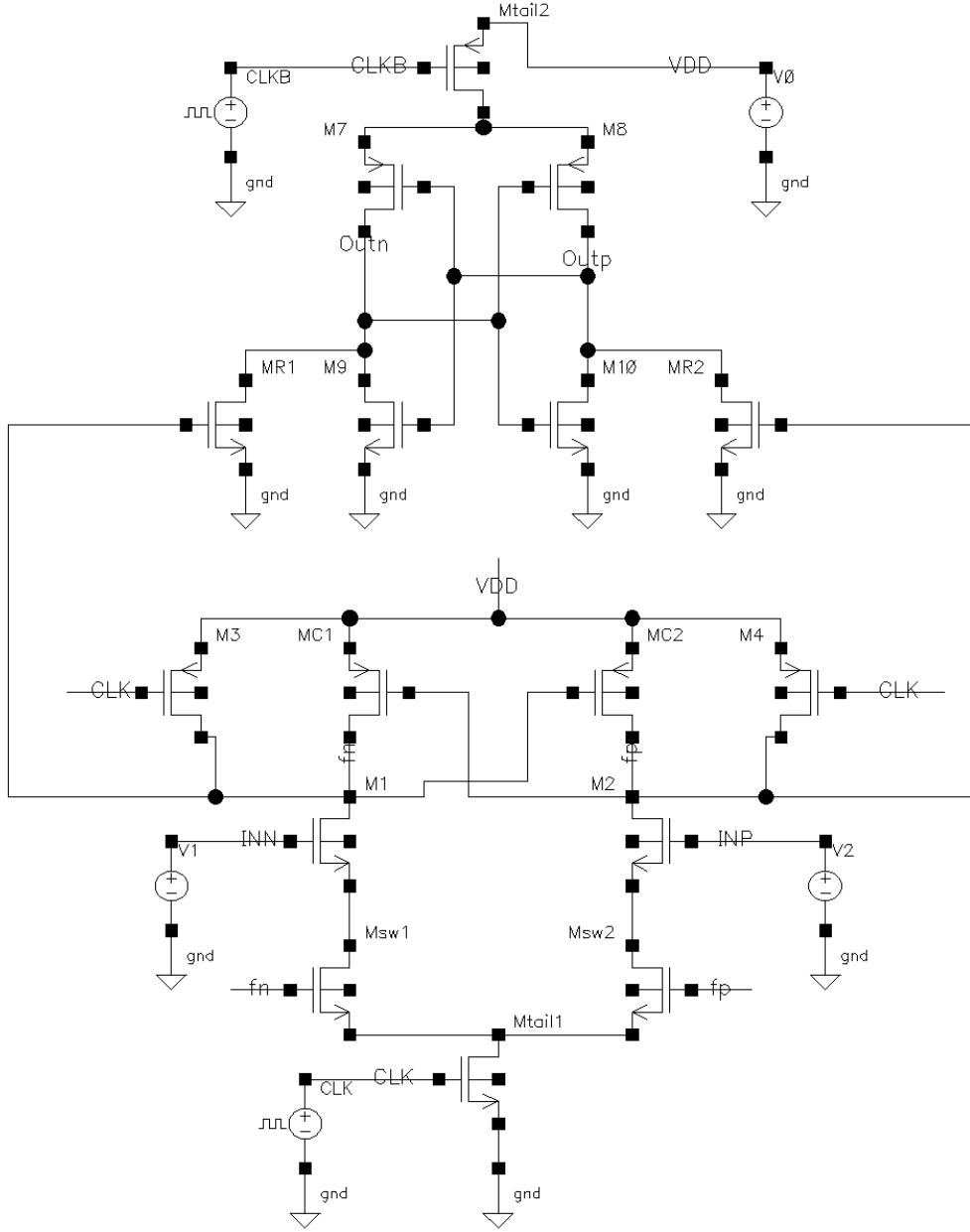


Figure 3.1: Schematic diagram of the proposed double-tail dynamic comparator

3.4 Working Principle

3.4.1 Reset Phase

When reset is taking place (when $CLK=0$), the tail transistor turns OFF, and at the same time, the internal nodes are charged up to V_{dd} levels. This is followed by the recharging of the output nodes to provide the correct initialization of the circuit to start the next evaluation stage.

During the evaluation stage, there is no static power consumption since there is no path between V_{dd} and ground.

3.4.2 Evaluation Phase

When the evaluation process is initiated (CLK=VDD), the tail transistors become active, and the internal nodes begin to discharge depending on the differential input voltage.

With the addition of the control transistors, once there's a difference on the nodes, positive feedback takes effect, resulting in faster discharges for one of the nodes while the other heads towards the power supply.

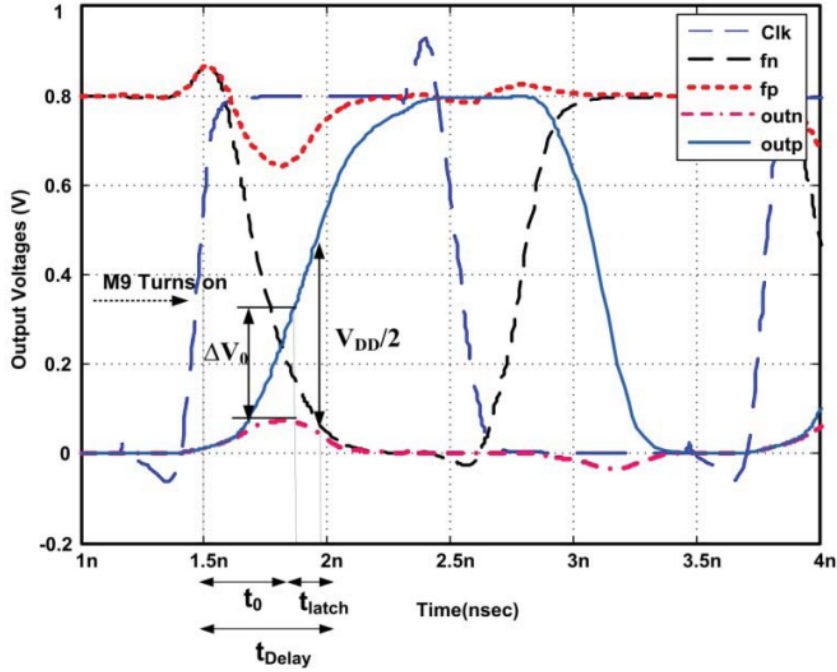


Figure 3.2: Transient response of the proposed double-tail comparator as reported in [1]

Figure 3.2 demonstrates the dynamic performance of the comparator under consideration for the case of short delays. In the evaluation phase, the internal nodes start discharging based on the differential input voltages leading to a delta voltage (ΔV_0) initially.

The additional control transistors further increase the voltage difference in order to enhance the positive feedback effect. Consequently, there will be earlier regenerations in the latch resulting in fast output switching when the node voltages exceed ($V_{DD}/2$).

Figure 3.3 displays the transient response from a Cadence Virtuoso simulation of the comparator. The clock triggers the switching from the reset phase to the evaluation one. When evaluation occurs, there will be a voltage difference generated between the two internal nodes (f_n and f_p), which will amplify through positive feedback effect.

The output signals switch their logic states rapidly meaning lower delay for the whole latch. The faster switching observed in the waveform clearly indicates lower delay in this design.

The transient waveforms in Fig. 3.2 and Fig. 3.3 show similar behaviors, confirming the

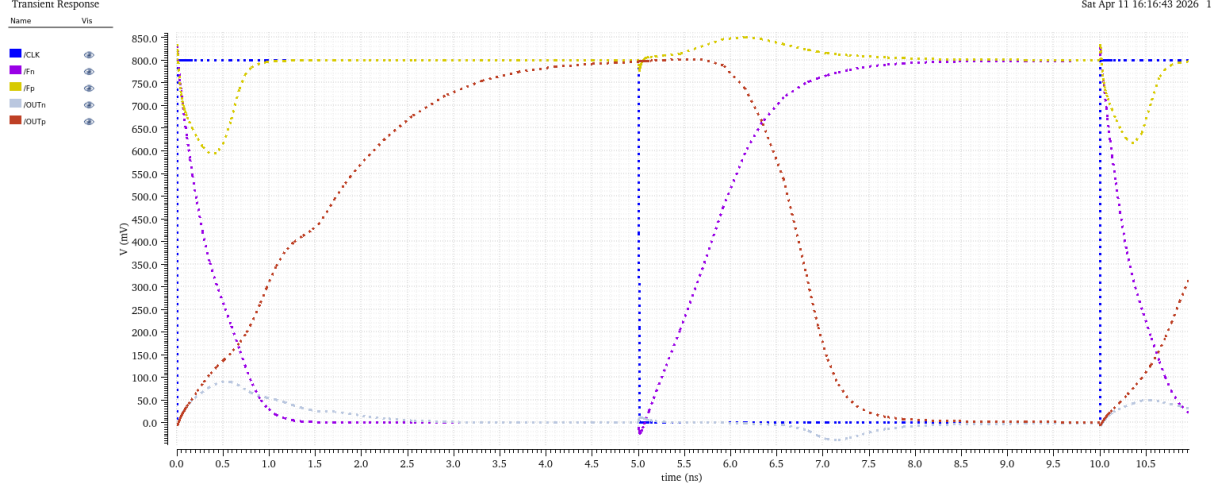


Figure 3.3: Transient response of the proposed comparator obtained using Cadence Virtuoso

correctness of the design implementation.

The simulation results indicate that the proposed dynamic comparator exhibits faster operation time with lower delay than other comparable structures.

3.5 Delay Analysis of the Proposed Comparator

3.5.1 Effect of Enhancing (ΔV_0)

The voltage difference (ΔV_0), which acts as an important parameter for determining the delay of the latch, is an essential consideration in the design. The inclusion of control transistors causes the value of (ΔV_0) to increase exponentially, resulting in reduced delay due to the shorter regeneration [1].

A larger (ΔV_0) results in faster amplification of the output voltage difference, thereby reducing overall delay.

$$\Delta V_0 \propto V_{in} \cdot e^{t/\tau}$$

3.5.2 Effect of Enhancing Transconductance

The effective transconductance of the latch $g_{m,\text{eff}}$ is increased by adding extra transistors within the improved version of the latch structure. This improves the positive feedback loop. The delay of the latch is reduced due to increased speed of regenerations.

$$t_{\text{latch}} = \frac{C_L}{g_{m,\text{eff}}} \ln \left(\frac{\Delta V_0}{V_{DD}} \right)$$

3.5.3 Reduction in Energy per Comparison

In the new comparator, energy savings are achieved as there is no need to charge and discharge internal nodes.

In contrast to conventional comparators, where all internal nodes switch at some point during the working process, the proposed comparator operates with a single node switching fully.

Chapter 4

Results and Discussion

4.1 Introduction

In this chapter, the proposed comparator design using double tails is simulated using Cadence Virtuoso simulator in order to study the practical performance of the proposed comparator as well as its capability to improve the speed of operation along with lowering down the power consumption.

The transient analysis, delay, and other challenges involved with such designs including mismatch and noise effects will be discussed in this chapter.

4.2 Performance of Proposed Comparator

The transient behavior of the comparator clearly indicates its functioning under different clock phases. When the clock is in the low state, the circuit is always under reset conditions and hence, all the internal nodes will begin from known initial states.

Once the clock reaches the high state, the evaluation process begins. Even the smallest difference in the input voltages causes a slight voltage difference to appear on the internal nodes of the comparator. This minute voltage difference is quickly amplified due to positive feedback action within the latch circuit.

As a result, one output node rushes to the supply voltage level whereas the other node tends towards ground voltage level.

4.2.1 Operation Analysis

Figure 4.1 shows the transient response for the suggested comparator, obtained through Cadence Virtuoso simulation. The operation of the circuit is controlled by a clock pulse and divides the time interval into reset and evaluation stages.

In the evaluation stage, an extremely low value of voltage difference generates a delta value (ΔV_0) at internal nodes.

As the amount of positive feedback increases, the difference gets magnified, and thus, there will be a swift switching of output nodes.

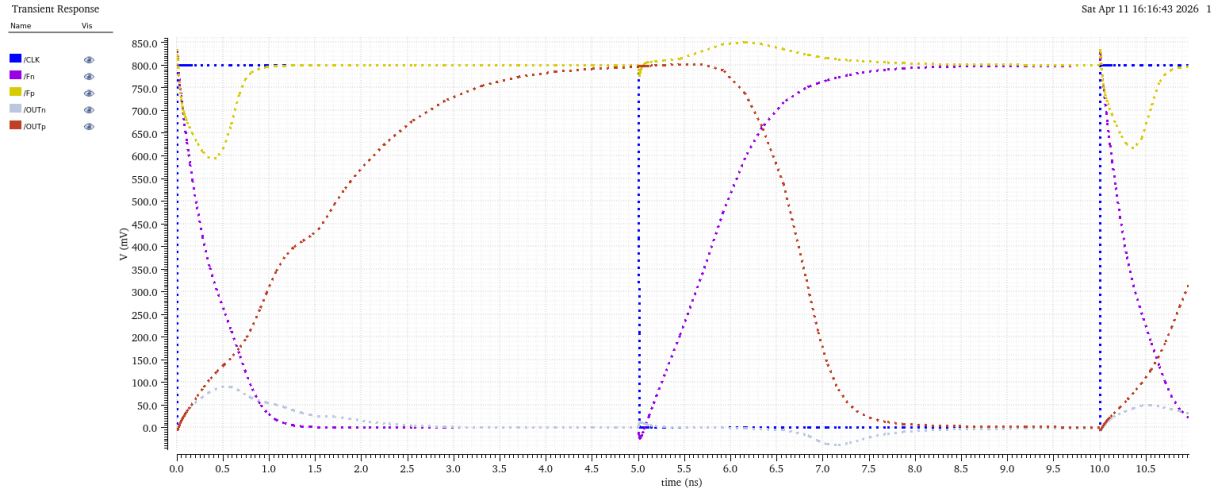


Figure 4.1: Transient response of the proposed comparator

4.2.2 Delay Analysis

The comparator delay is one of the most important parameters, determining the speed of response from the circuit to input signal changes. The delay consists of two major factors: the time necessary for the internal nodes to start diverging and the time required by the latch to amplify the differences.

In our circuit, the delay decreases since there is an increased voltage difference between the internal nodes initially, enabling faster response from the latch. Also, the latch gain increases, facilitating the regenerative process.

Due to these modifications, the outputs toggle faster, thus reducing the comparator delay.

4.3 Design Considerations

4.3.1 Mismatch Analysis

When implemented in practice, the two transistors used in the circuit will not be exactly the same. This results in a slight variation in the output voltage, hence affecting the accuracy of the switching points set for the comparator.

To overcome this problem, good design techniques ensure that the transistors have the right dimensions and symmetry. It is impossible to remove the mismatch completely.

4.3.2 Kickback Noise

Another important factor to consider while designing dynamic comparators is kickback noise. Kickback noise results due to the coupling of output transitions into the input stage through parasitic capacitances causing disturbance to the input signals.

In the new design, the regeneration effect makes the circuit operate faster but at the cost of

increased kickback noise. Thus, appropriate isolation techniques must be adopted during the layout phase to reduce this effect.

4.4 Comparison with Existing Designs

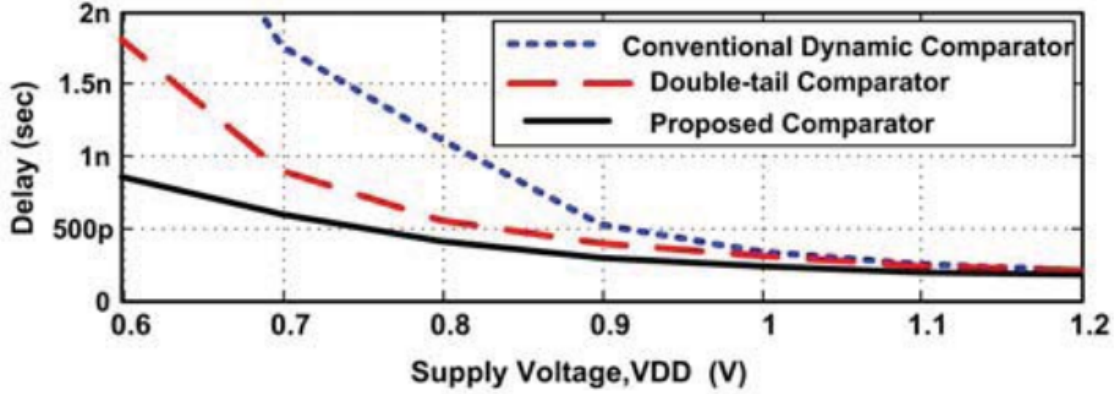


Figure 4.2: Delay comparison of comparators [1]

Figure 4.2 depicts the delay variation concerning the supply voltage for the various comparator configurations. With an increase in the supply voltage, the delay decreases.

Due to stronger positive feedback and the initial voltage difference, the proposed comparator demonstrates faster operation compared to standard and double-tail comparators. It makes the proposed comparator a good choice for low-voltage operation.

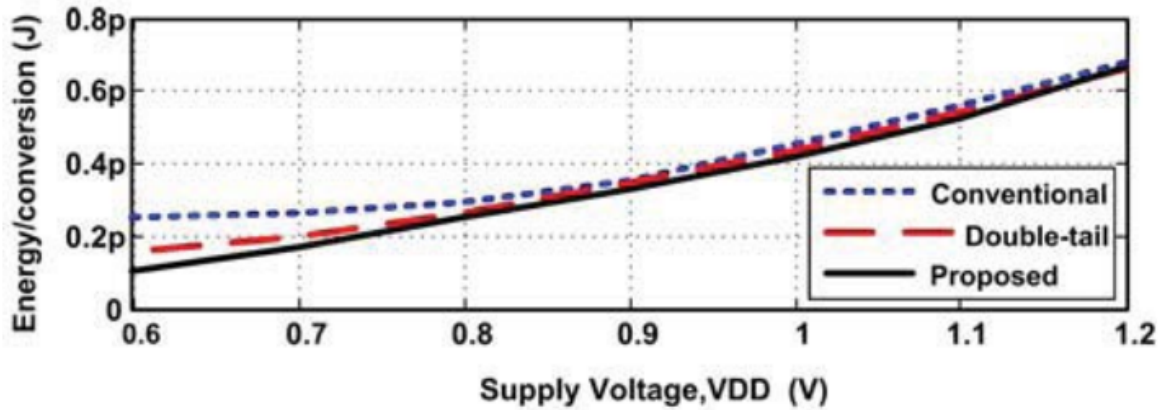


Figure 4.3: Energy per conversion comparison [1]

The energy consumption in the proposed comparator is lower compared to the standard comparator because of lesser switching events and regeneration.

In general, the proposed comparator provides superior performance concerning speed and energy consumption; hence, it can operate at higher frequencies with lesser delays compared to the standard and double-tail comparators.

Although the area of the proposed comparator is greater than the standard comparator, the advantages in speed and energy efficiency make it suitable for modern applications.

The values in the table are taken from previously reported results for comparison purposes.

Table 4.1: Performance Comparison

Comparator	Delay	Power	Speed
Conventional	High	Low	Moderate
Double-Tail	Medium	Medium	High
Proposed	Low	Low	Very High

Table 4.2: Performance Comparison of Different Comparators

Parameter	Conventional	Double-Tail	Proposed
Technology	180 nm	180 nm	180 nm
Supply Voltage	0.8 V	0.8 V	0.8 V
Max Frequency	900 MHz	1.8 GHz	2.4 GHz
Delay	High	Medium	Low
Energy per Conversion	0.3 pJ	0.27 pJ	0.24 pJ
Offset Voltage	7.89 mV	7.91 mV	7.8 mV
Area	Small	Medium	Medium

Performance comparison Table 4.1 makes it quite evident that the suggested comparator surpasses the classic as well as the double-tail designs. Though being economical in terms of power, the former suffers greatly in terms of time taken to perform operations due to its delayed functioning. In order to overcome such delay, the double-tail design uses the approach of minimizing transistor stacking, but even then it is not able to reduce either delay or power consumption considerably. However, in case of the proposed comparator, low delay and power consumption go hand-in-hand to increase its speed.

The details of the comparison are shown in Table 4.2 using the basis of the design parameters. All the comparators have been designed using the same technology node of 180 nm and use the same supply voltage of 0.8 V to make the comparison credible. The proposed comparator is the winner in terms of operating frequency at 2.4 GHz and also boasts reduced delays and less power dissipation compared to other comparators. Although all the designs offer comparable offset voltages and occupy similar chip area, the proposed design offers a better trade-off between speed and efficiency.

Chapter 5

Conclusion

5.1 Conclusion

In this paper, dynamic comparators are thoroughly examined and compared against their conventional counterparts, double-tailed comparators, as well as their enhancements.

Areas where issues of delay and low-voltage operation present difficulties are highlighted. Based on the knowledge gained, an enhanced double-tailed dynamic comparator has been designed using Cadence Virtuoso software.

The proposed design improves the operation of the dynamic comparator through increased voltage difference across the two nodes and improved feedback. From simulations carried out, it is evident that the enhanced design offers reduced delays as opposed to conventional dynamic comparators without consuming more power.

This makes it ideal for applications that require high speed with reduced power consumption.

5.2 Future Work

Although the proposed comparator already outperforms the existing comparator in terms of speed and energy consumption, there are still some interesting paths that can be pursued.

One such direction is lowering the threshold voltage for certain nodes in order to increase speed while simultaneously reducing energy consumption.

Another good idea might be implementing a particular design of the comparator, followed by simulations in order to evaluate the impact of the parasitic effects on its performance.

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